

Polynomial Regression for Geothermal Systems

Machine Learning Assignment 1

Roll number: BT2024128
Problems: Steam turbine optimization (var1), thermal reservoir mapping (var2)
Primary metrics: Mean Squared Error (MSE), coefficient of determination (R^2)

Abstract

Two polynomial regression models were developed for the personalized datasets assigned to BT2024128. Feature subsets were screened with ordinary least squares (OLS), then polynomial degree and Ridge regularization strength were compared using five-fold cross-validation on the training data. Polynomial expansion and scaling were fitted within each validation fold. The selected var1 model uses all six inputs, degree 5, and Ridge penalty $\alpha = 10$; the var2 model uses all three inputs, degree 12, and $\alpha = 1$. Repeated five-fold cross-validation produced MSE 0.4378 and $R^2 = 0.9593$ for var1, and MSE 0.2291 and $R^2 = 0.9945$ for var2.

1 Dataset and task

Each problem contains 1,000 training observations and 1,000 test observations. All columns are numeric, all input variables lie between -1 and 1 , and no values are missing. The training sets contain the continuous target y ; the test targets are hidden.

Problem	Training shape	Test shape	Candidate inputs
var1	1000×7	1000×6	$x_1, \dots, x_6$
var2	1000×4	1000×3	x_1, x_2, x_3

The train and test files for each problem were kept separate. Test values were used only after the final configuration had been selected and fitted on the complete corresponding training set.

2 Polynomial regression pipeline

For p selected inputs and maximum degree d , the expansion contains every monomial whose total degree is at most d . Excluding the constant term, the number of generated terms is

$$N(p, d) = \binom{p+d}{d} - 1.$$

The prediction model is linear in these expanded terms:

$$\hat{y} = \beta_0 + \sum_{j=1}^{N(p,d)} \beta_j \phi_j(\mathbf{x}).$$

Each final scikit-learn pipeline applied `PolynomialFeatures`, then `StandardScaler`, followed by `Ridge`. Its coefficients minimize

$$\sum_i (y_i - \beta_0 - \mathbf{z}_i^\top \boldsymbol{\beta})^2 + \alpha \|\boldsymbol{\beta}\|_2^2,$$

where $\mathbf{z}_i$ contains the standardized polynomial terms. Ridge shrinks large coefficients and helps control variance when many interaction terms are present. The intercept is not penalized. Keeping polynomial expansion and scaling inside the pipeline prevents preprocessing leakage across validation folds.

3 Model-selection procedure

The following procedure was applied independently to var1 and var2.

1. Screen every non-empty feature subset using OLS polynomial regression. Degrees up to 10 for var1 and 20 for var2 were considered, omitting OLS designs with more than 700 terms against 800 fitting observations per fold.
2. Keep the full feature sets because they had the lowest OLS validation error by a large margin.
3. Search Ridge models on these feature sets: degrees 3–6 and $\alpha \in \{0.1, 1, 3, 10, 30, 100\}$ for var1; degrees 7–16 and $\alpha \in \{0.01, 0.1, 1, 3, 10\}$ for var2.
4. Compare MSE (primary) and R^2 (secondary) using shuffled five-fold cross-validation with random seed 42. Choose the lowest mean MSE.
5. Re-evaluate each chosen configuration using five repeats of five-fold cross-validation.

Ridge permits more polynomial terms than OLS because the positive penalty stabilizes coefficient estimation. The Ridge search was focused around degrees where the OLS screen showed useful nonlinear structure. The held-out test targets were never used in these decisions.

3.1 Degree comparison

Table 1: Five-fold cross-validation of representative configurations

Problem	Model	Degree	α	Terms	Mean MSE
var1	OLS	4	0	209	0.7725
var1	OLS	5	0	461	1.0671
var1	Ridge	5	3	461	0.4731
var1	Ridge	5	10	461	0.4265
var1	Ridge	5	30	461	0.4408
var2	OLS	8	0	164	0.2544
var2	Ridge	10	1	285	0.2346
var2	Ridge	12	1	454	0.2268
var2	Ridge	13	1	559	0.2307

For var1, degree 5 OLS overfits, but Ridge at the same degree reduces held-out error sharply. For var2, the best Ridge result occurs at degree 12. Degrees above that did not improve the five-fold MSE. The tested penalty values also show why the selected α values are reasonable rather than arbitrary.

4 Feature selection

The preliminary OLS screen favored all available inputs for both datasets. The best five-feature var1 alternative (x_1, x_2, x_3, x_5, x_6 , degree 3) had MSE 1.8294, compared with 0.7725 for the full-feature degree-4 OLS baseline. All lower-dimensional var2 subsets were far behind the full three-feature degree-8 OLS model. Thus Ridge tuning was performed on the full feature sets; interactions across the variables were retained.

5 Final validation results

Table 2: Repeated five-fold validation of the selected models (five repeats)

Problem	Features	Degree	α	Terms	MSE	R^2
var1	$x_1, \dots, x_6$	5	10	461	0.4378 ± 0.0502	0.9593 ± 0.0063
var2	x_1, x_2, x_3	12	1	454	0.2291 ± 0.0251	0.9945 ± 0.0013

The repeated estimates are close to the original five-fold screening scores, which shows that the choices are not tied to one random split. A separate five-fold out-of-fold prediction run gave the following results.

Problem	OOF MSE	OOF R^2	Training MSE	Training R^2
var1	0.4265	0.9608	0.1602	0.9853
var2	0.2268	0.9947	0.1487	0.9965

5.1 Diagnostic interpretation

For var1, out-of-fold predictions track the identity line across negative and positive scores. Residuals are centered near zero, with a few larger deviations at extreme responses. Ridge controls the variance of the 461-term degree-5 expansion; without regularization, its five-fold MSE was 1.0671.

For var2, out-of-fold predictions lie close to the identity line over the observed target range, approximately -30 to 39 . Residuals remain centered around zero without a strong systematic curve. Degree 12 captures more structure than the earlier degree-8 OLS baseline while Ridge limits unstable coefficients.

6 Final training and test inference

After selection, each pipeline was refitted on all 1,000 observations from its training dataset. Predictions were generated in the original test-row order. The final output ranges were -11.7862 to 15.1081 for var1 and -29.7657 to 38.7586 for var2. Both submission files follow the provided sample: exactly 1,000 finite predictions in a single column named `y`, with no exported row index.

7 Reproducibility and safeguards

The implementation provides separate scripts for selection and inference. The selection script writes OLS and Ridge search tables, repeated-validation metrics, a degree-selection figure, and out-of-fold diagnostic plots. The prediction script validates schemas, rejects missing or nonnumeric values, checks prediction length and finiteness, saves the fitted pipelines, and writes the two submission CSV files.

- **No test leakage:** Neither test inputs nor hidden targets influence feature or degree selection.
- **Pipeline isolation:** Polynomial generation and scaling are refitted inside every validation fold.
- **Reproducible splits:** All shuffled validation procedures use random seed 42.
- **Complexity control:** OLS screening avoids expansions larger than 700 terms; Ridge allows larger designs with coefficient shrinkage.
- **Submission checks:** Output row counts, column name, order, and finite numeric values are verified programmatically.

8 Limitations

The true test targets are hidden, so expected performance is estimated from cross-validation rather than directly measured. Polynomial regression can also become sensitive near the edges of the input domain, particularly for the six-variable var1 model. The supplied train and test inputs share the same bounded range of $[-1, 1]$, which limits extrapolation risk, but some test points combine boundary values more frequently than the training data. Predictions at those combinations may extend beyond the observed training-target range.

Cross-validation is an estimate, not a measurement against the hidden test targets. Because the same training set informed the model search and reported fixed-model validation, the repeated scores may be mildly optimistic. If additional labeled data became available, the most useful next check would be a final untouched holdout assessment.

9 Conclusion

Cross-validated selection produced two reproducible polynomial models with Ridge regularization. Degree 5 with $\alpha = 10$ gives var1 substantially lower held-out error than the OLS baseline. Degree 12 with $\alpha = 1$ provides a smaller but consistent gain for var2. Both pipelines were retrained on all available labeled observations and used to create the final prediction files for BT2024128.